

Clinical and Oral Microbiology

Eukaryotic Pathogens

Integrated Biomedical and Clinical Sciences OMFS 513

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Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Objectives

Learning Outcomes

1. Describe the classification of fungal family.
2. Describe the pathogenesis different types of fungal infections, their oral manifestations and symptoms.
3. Describe antifungal agents used to manage candidiasis.
4. Describe the basis of parasitic infections and general treatment guidelines.

Outline

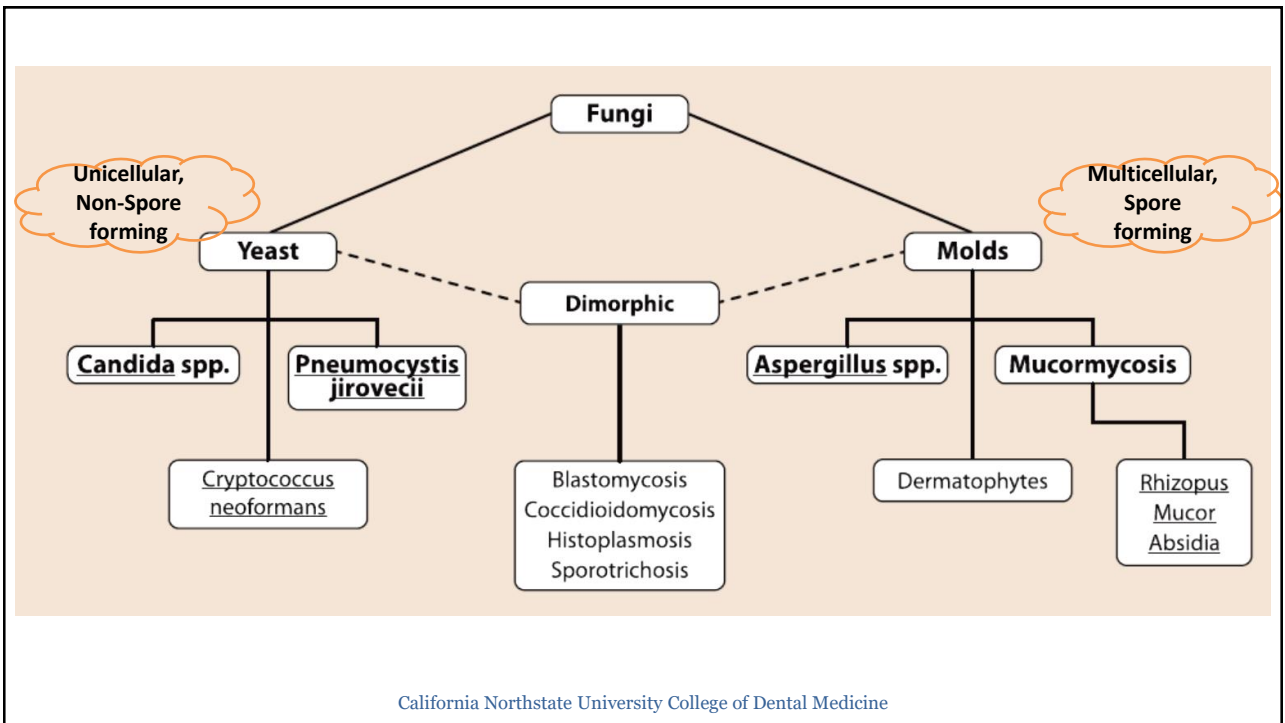
- 1- Classification.
- 2- Forms Fungal infections.
- 3- *Candida sp.*, virulence, symptoms and oral manifestation.
- 4- Symptoms and oral manifestations of different fungal infections.
- 6- Anti-fungal medications
- 7- Parasites and treatment strategies.

Kingdom Fungi

Myceteae

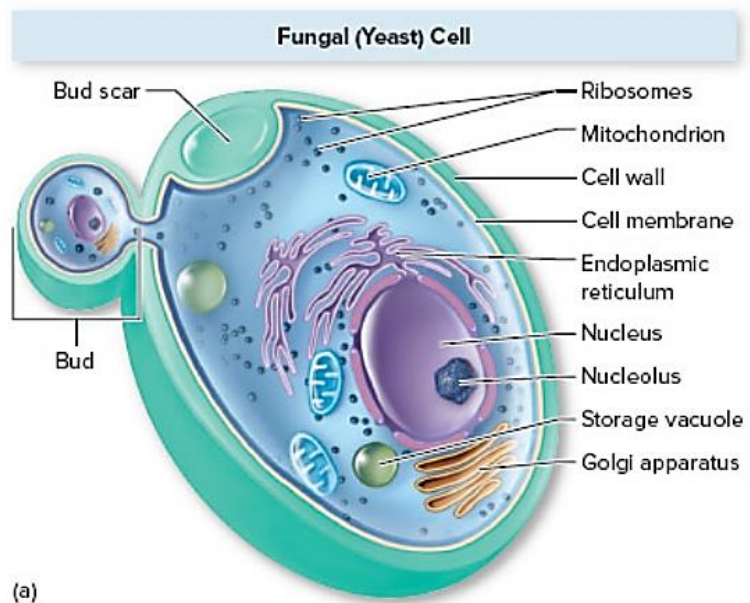
Macroscopic like
mushrooms, puffballs.

Microscopic like yeast
mold & dimorphic



Yeasts:

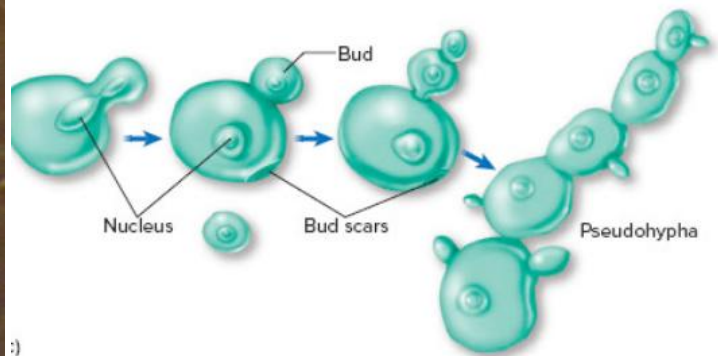
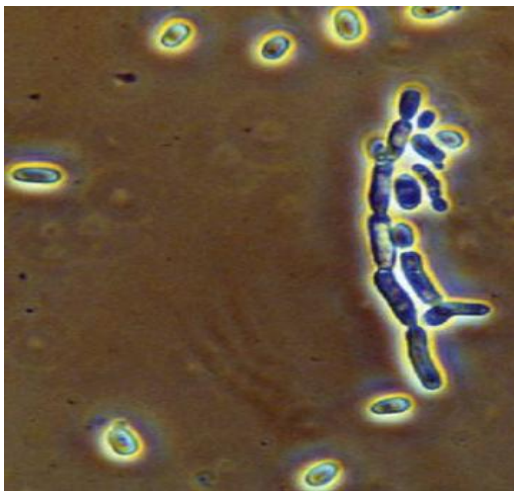
- Single-celled fungi
- Generally oval shaped.
- Capable of sexual and asexual reproduction.
- Asexual by budding.



[VitalSource Bookshelf: Oral Microbiology and Immunology](#) California Northstate University College of Dental Medicine

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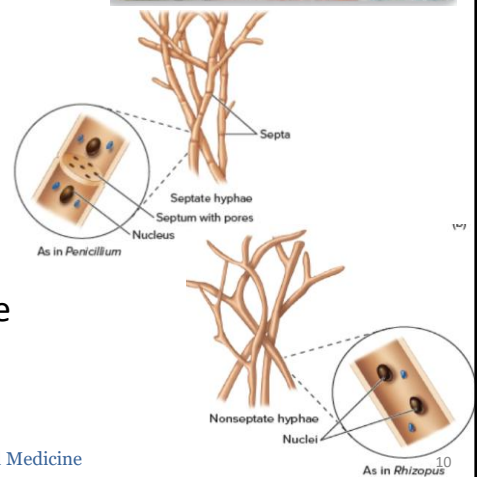
In some cases, the daughter buds do not separate (e.g: *Candida albicans*) resulting in a pseudohyphae formation.



Pseudohyphae: *Candida albicans* buds fail to separate and continue elongation.

Mold

- Multicellular, filamentous fungi.
- Structural unit: Hyphae.
- Hyphae long filaments that can have:
 - Septate structure
 - Non-septate structures.
- Reproduction sexual & asexual via spores
 - Most common: Conidiospores (aka, conidia or free spores. Conidia structure vary by the species.
 - Enclosed in the sporangium (spore bearing sac)
- Spore-inhalation is a recognized form of disease transmission.



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Dimorphic:

- Dimorphic fungi can switch between the yeast and mold states according to the environmental conditions like temperature, PH, etc.
- Mold form in environment.
- Yeast form in susceptible host.
- Examples include: *Candida albicans*, *Blastomyces*, *Coccidioides*, *Histoplasma*.

Distinctive features

1- Eukaryotes, heterotrophic, saprobes, free-living, strict aerobes, capable of sexual and asexual reproduction.

2- Cell wall: antigenic, protective and determine cell shape.

- The cell wall is composed of complex carbohydrates like chitin with glucans and mannose-proteins. Glucan (polysaccharide not found in humans) is the antifungal target of the echinocandins like caspofungin.

3- Cell membrane: “Ergosterol” is the dominant fungal membrane sterol (drug target).

Fungal diseases

A. Fungal allergies

Due to spore inhalation. Examples include farmer's lung due to allergic bronchopulmonary aspergillosis. High anti-aspergillosis IgE leads to formation of mucus plug and asthma-like symptoms.

B. Mycotoxicosis

Due to ingestion of fungal toxins. For example:

- Ergotism can result from ingesting ergot-contaminated rye bread (convulsion, hallucinations).
- Aflatoxin from *aspergillus sp.* contaminate grains and release aflatoxin. Aflatoxin B1 can cause loss of function mutation in *P53* gene and liver cancer.

C. Mycoses

- Caused by opportunistic or pathogenic fungi.
- Most pathogenic fungi are exogenous (natural habitats water, soil, and organic debris).
- Highest incidence: candidiasis and dermatophytosis.
- Infections range from cutaneous to systemic infections that can be fatal in the immunocompromised hosts.

GENUS	TYPICAL GROWTH
Superficial Fungi	
<i>Epidermophyton</i>	Mold
<i>Microsporum</i>	Mold
<i>Trichophyton</i>	Mold
Subcutaneous Fungi	
<i>Sporothrix</i>	Dimorphic
Opportunistic Fungi	
<i>Aspergillus</i>	Mold
<i>Candida</i>	Dimorphic
<i>Mucor</i>	Mold
<i>Rhizopus</i>	Mold
<i>Pneumocystis</i>	Cysts ^b
Systemic Fungi	
<i>Blastomyces</i>	Dimorphic
<i>Coccidioides</i>	Dimorphic
<i>Histoplasma</i>	Dimorphic
<i>Cryptococcus</i>	Yeast

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Types of Mycosis

1- Superficial

2- Cutaneous/mucocutaneous mycosis

Dermatophytes causing tinea, Candidal infection,

3- Sub-cutaneous mycosis:

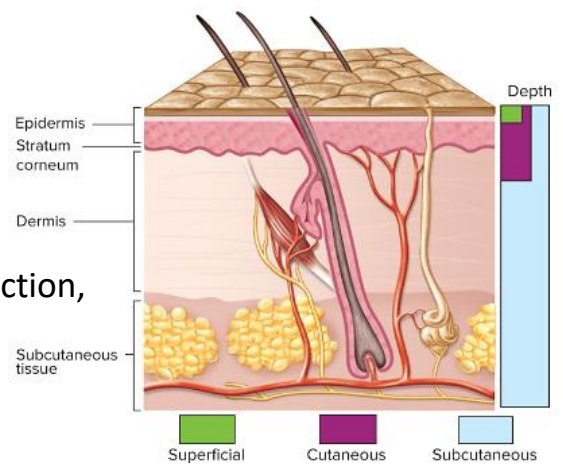
Sporothrix causes rose-gardener's disease.

4- Systemic mycosis "deep mycosis":

a) **Pathogenetic:** usually in specific geographic areas. Examples include Histoplasmosis, Blastomycosis, Coccidioidomycosis

b) Opportunistic

Candida, Aspergillus, Cryptococcus, Mucormycosis, Pneumocystis



Diagnosis

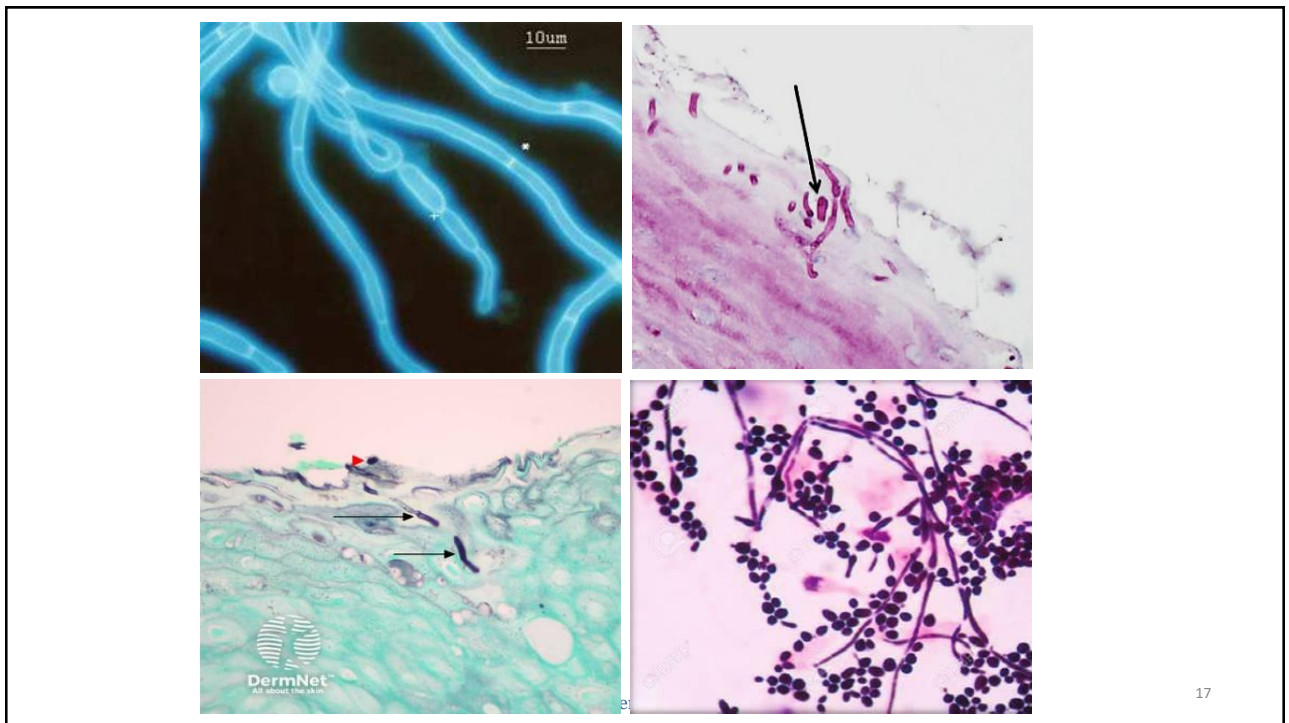
- **Clinical presentation**

- **Microscopic examination:**

Clinical specimen + a 10% solution of potassium hydroxide (KOH) to digest human tissue. Observe fungi under microscope with/without staining.

Stains include:

- **Calcofluor white** stains fungal chitin (fluorescent blue on black background).
- **PAS staining:** fungi appear pink.
- **Gomori methenamine silver stain (GMS).** Fungi appear brownish black.



- **Culture:** slow growth and require special media. Example: Sabouraud's agar (high carbohydrate media)
- **Detection of nucleic acid**
- **Sereology:** Detection of fungal antigens or host-antibodies.

Pathogenesis

Adherence

Adherence mediated by fungal adhesins and host cell receptors

Invasion

Fungi access the body through skin breaks or via inhalation. Invasion across mucosal barriers may involve enzymatic digestion of the ECM. Dimorphic fungi can change their morphology to allow efficient tissue invasion.

Injury

Either attributed to fungal damage to the surrounding tissue or collateral damage from the immune system. The response to infection with several systemic fungi is the formation of granulomas (e.g., coccidioidomycosis, histoplasmosis, and blastomycosis). Fungi has no endotoxins or exotoxins. Some species have mycotoxins

Skin: Cutaneous/subcutaneous infections

Mucocutaneous infections

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Cutaneous Infections

- Dermatophytosis (aka, tinea) caused by dermatophytes from genus *Trichophyton*, *Epidermophyton*, and *Microsporum*.
- **Transmission:** from soil, direct contact with infected persons or pets.
- All three dermatophytes are molds affect the epidermis bellow the keratin layer but above the basal layer.
- Can affect skin, hair and nails.

[Tinea Capitis: Symptoms, Causes & Treatment \(clevelandclinic.org\)](#)

[California Northstate University College of Dental Medicine](#)

[Cutaneous & Subcutaneous Mycoses | Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, 17e | AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#)

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Presentation

- **Body: Tinea corporis**
 - Lesions are itchy, scaly, well-circumscribed, hyperpigmented, usually inflamed circular borders with vesicles surrounding a clear area of relatively normal skin or central scaling.
- **Foot: Tinea pedis “athlete's foot”.**
 - Most prevalent.
 - Present by itching between the toes and small vesicles that rupture and discharge a thin fluid. Chronic: pain and pruritus, and cracks which can predispose to secondary bacterial infection.



Athlete's Foot



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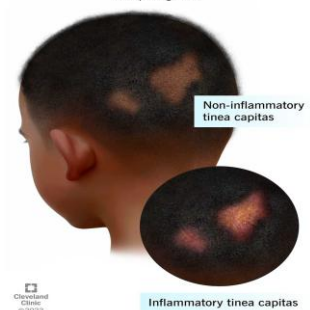
- Nail: Onychomycosis.
 - Nails lose the natural luster become yellow, brittle, thickened, and crumbly. One or more nails of the feet or hands may be involved.
- Head: Tinea Capitis
 - Infection produces dull gray, circular patches of alopecia, scaling, and itching.

Management:

Dermatophytosis: topical antifungal like imidazole except nail infection, long-term oral antifungals (-azoles)



Tinea Capitis
Scalp ringworm



Non-inflammatory
tinea capitis

Inflammatory tinea capitis

Subcutaneous Infections

- Sporotrichosis (rose gardener's disease) defines ulcers on skin (most common) that presents as red or pink nodules that take too long to heal.
- Caused by *Sporothrix schenckii*: dimorphic fungus "cigar-shaped" fungus.
- **Transmission:** Grow in the soil and is introduced to the body through open wounds or thorn injury.

- About 75% of cases are lymphocutaneous.
- Infection is subcutaneous that spread to lymphatics yielding granulomatous nodules w ulceration.
- Immunocompromised patients: can cause disseminated sporotrichosis.

Management: Itraconazole



[Cutaneous & Subcutaneous Mycoses | Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, 17e | AccessMedicine | McGraw Hill Medical | cnsu.edu](#)



Emerging fungus “*Candida auris*”

- *C. auris* is an emerging fungus. Long survival on fomites.
- First reported in Japan in 2009. Began spreading in the US since 2016. 4500 cases reported in 2023. mortality rate, around 30 %.
- *C. auris* mostly affects patients with underlying medical conditions. Can spread in healthcare setting and nursing homes.
- Cause multiple conditions including candidemia (associated with poor prognosis and high mortality rate, up to 60 %).
- Resistance to several antifungal groups. Can respond to medication of the echinocandins group (e.g: Caspofungin; inhibit β -glucan & cell wall synthesis).

[About Candida auris \(C. auris\) | Candida auris | Fungal Diseases | CDC](#)

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Candidiasis

“Disease of the Diseased”

- *Candida albicans* is presents in the oral cavity of healthy individuals.
- Also present in GI tract, female genital tract and sometimes the skin.
- Characters: Oval budding yeast that can form pseudohyphae.
- *Candida* can cause oral mucosal diseases.

TABLE 8.1

Fungal species recovered from the human mouth

<i>Candida</i> species	Other fungal species (rare)
<i>Candida albicans</i>	<i>Paracoccidioides brasiliensis</i>
<i>Candida glabrata</i>	<i>Aspergillus</i> spp.
<i>Candida tropicalis</i>	<i>Cryptococcus neoformans</i>
<i>Candida krusei</i>	<i>Histoplasma capsulatum</i>
<i>Candida lusitanae</i>	<i>Mucor</i> spp.
<i>Candida dubliniensis</i>	<i>Saccharomyces</i> spp.
<i>Candida kefyr</i>	<i>Geotrichum</i> spp.
<i>Candida guilliermondii</i>	<i>Rhizopus</i> spp.
<i>Candida parapsilosis</i>	

Risk factors

- Older age
- Immunocompromised status
- Malnutrition and vit. deficiency
- Certain medications like antibiotics, corticosteroid and immunosuppressives.
- Uncontrolled diabetes
- Xerostomia

Age: being very young or old

HIV infection

Mucosal trauma/irritation, including denture wearing

Drug therapy

Antibiotics

Corticosteroids (local or systemic)

Immunosuppressives

Cytotoxics

Malnutrition

Iron deficiency

Vitamin B₁₂ deficiency

Undiagnosed or poorly controlled diabetes mellitus

Leukemia

Agranulocytosis

Xerostomia

High-carbohydrate diet

[VitalSource Bookshelf: Oral Microbiology and Immunology](#)

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Virulence factors associated with *Candida albicans*

A) Host-related factors

Several factors including changes in the host's immune response or changes in the composition of the oral microbiota.

B) *Candida* virulence factors

1) Adherence

Increased adherence allows *Candida* to resist removal by salivary flow and swallowing. *Candida* has several cell surface adhesins to promote pathogen adherence.

Example: agglutinin-like sequence (ALS) proteins and Hyphal wall protein 1 (Hwp1)}.

2) Evasion of host defenses/Destruction of host tissue.

i) *C.albicans* can switch from its yeast morphology to a filamentous form (hyphae) which facilitates epithelial penetration and increase the resistance to phagocytosis. Transformation to the hyphae forms can be triggered by environmental factors, limited availability of certain nutrients & genetic factors.

ii) Phenotypic Switching: *C. albicans* can rapidly change its cell morphology which will affect the surface antigenicity, adhesiveness, and phagocytosis resistance.

iii) Hydrolytic Enzymes: Release of extracellular hydrolytic enzymes can facilitate tissue damage. Examples: aspartyl proteinases (SAPs) and phospholipases.

Virulence factor	Effect
Adherence	Promotes retention in the mouth
• Relative cell surface hydrophobicity	• Non-specific adherence process
• Expression of cell surface adhesin molecules	• Facilitates specific adherence mechanisms
Evasion of host defences	Promotes retention in the mouth
• High frequency phenotypic switching	• Antigenic modification through frequent cell surface changes
• Hyphal development	• Reduces likelihood of phagocytosis; allows phagocytosed yeast to escape phagocyte
• Secreted aspartyl proteinase production	• Secretory IgA destruction
• Binding of complement molecules	• Antigenic masking
Invasion and destruction of host tissue	Enhances pathogenicity
• Hyphal development	• Promotes invasion of oral epithelium
• Secreted aspartyl proteinase production	• Host cell and extracellular matrix damage
• Phospholipase production	• Damage to host cells

Candidiasis

- A- Pseudomembranous candidiasis.
- B- Acute erythematous candidiasis.
- C- Chronic hyperplastic candidiasis.
- D- Chronic erythematous candidiasis.

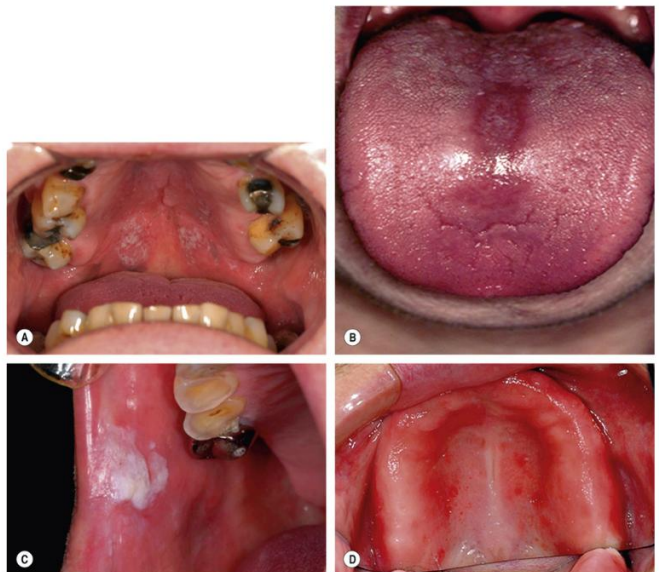


FIGURE 8.2 Clinically distinct forms of oral candidosis (A) pseudomembranous candidosis (thrush), (B) acute erythematous candidosis, (C) chronic hyperplastic candidosis, (D) chronic erythematous candidosis.

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VitalSource Bookshelf: Marsh and Martin's Oral Microbiology Northstate Ur

1- Pseudomembranous Candidiasis

Also termed oral thrush. It is characterized by white plaque-like lesions on the oral mucosa that can be removed by gentle scraping to reveal an underlying erythematous mucosa.

2- Acute Erythematous Candidiasis

It manifests as painful, reddened patches on the oral mucosa. This form is primarily associated with prior use of broad-spectrum antibiotic, especially if the patient was also using steroid inhaler.

3- Chronic Hyperplastic Candidiasis

This is a comparatively rare form of oral candidiasis. It can be observed in middle-aged males with history of smoking and high rate of alcohol consumption. It is characterized by Candida hyphal invasion of the oral epithelium. Lesions are most frequently evident as bilateral white patches in the buccal commissure regions of the mouth. **Those lesions cannot be removed by gentle scraping without bleeding.** Chronic hyperplastic candidiasis can be associated with the development of oral cancer at the lesion site.

4- Chronic Erythematous Candidiasis

This is also termed candida-associated denture stomatitis. This infection is seen as reddening (erythema) of the mucosa beneath the fitting surface of a denture. Chronic erythematous candidiasis can develop under acrylic dentures or intra-oral appliance. It is mostly observed on the palate rather than the mandibular mucosa. This condition can be caused by inadequate oral hygiene or poor-fitting denture. Invasion of the underlying epithelium is not observed in chronic erythematous candidiasis.

Other forms of Candidiasis



FIGURE 8.4 *Candida*-associated lesions as illustrated by angular cheilitis.

Angular Cheilitis

[Median Rhomboid Glossitis: A Clinical and Microbiological Study - PMC \(nih.gov\)](#)

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Figure 1

The appearance of median rhomboid glossitis (MRG).



Median Rhomboid Glossitis

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1- **Angular cheilitis** presents as erythematous lesions at the corners of the mouth. The infection can be observed with other forms of oral candidiasis, particularly chronic erythematous candidiasis. Angular cheilitis can also be attributed to mixed infections with *Staphylococcus aureus* or *streptococci*.

2- **Median Rhomboid Glossitis (Central Papillary Atrophy),**

Median rhomboid glossitis is a chronic infection presented as atrophy of the filiform papillae with a red area in the central dorsum of the tongue. The infection is strongly associated with smoking and use of inhaled steroids. Median rhomboid glossitis can be symptomatic (presenting with a discomfort or a burning sensation). A “kissing lesion” can develop where there is a corresponding area of erythematous candidiasis on the hard palate.

3- **Chronic Mucocutaneous Candidiasis**

This disease is manifested by *Candida* infections in the skin, mucous membranes and nails. Chronic mucocutaneous candidiasis can be attributed to a rare congenital condition that leads to impaired cellular immunity against *Candida*. Affected patients may also present with endocrine disorders.

Isolation of *Candida* from the oral cavity

Isolation method	Advantages	Disadvantages
Culture of whole saliva	Sensitive; viable organisms isolated	Problems may occur with collection of sample; not site specific
Concentrated oral rinse	Quantitative; viable cells isolated	Some patients have difficulty in using rinse; not site specific
Swab	Simple to use; viable cells isolated; site specific	Difficult to standardise
Smear	Simple to use; not reliant on culture	Viable cells not determined; species identity not readily confirmed
Imprint culture	Quantitative; viable cells isolated; site specific	Some sites difficult to sample
Biopsy	Essential for chronic hyperplastic candidosis	Invasive; not appropriate for other forms of candidosis

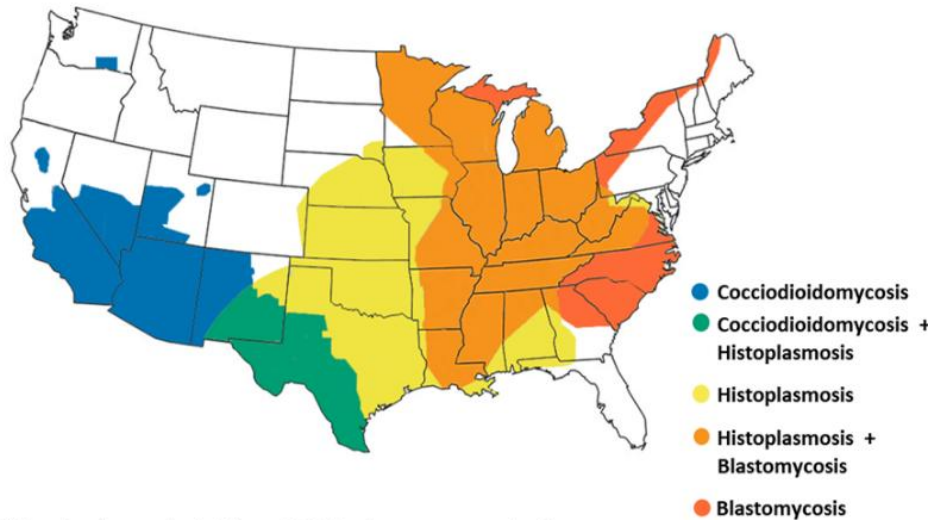
Deep Fungal Infections

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Deep Fungal Infections

- **Where is the organisms found.**
 - Geographic location
 - Animal droppings
 - Nosocomial infections
 - Food stuff
- **Type of infection**
 - Acute
 - Chronic (mimics symptoms seen in tuberculosis)
 - Disseminated (hematogenous or lymphatic spread)
 - Immunocompromised patients
- **Clinical presentation**
 - Oral infections can present as non-healing ulcers, granular growths, or palatal perforations.

Deep Fungal Infections - Locations

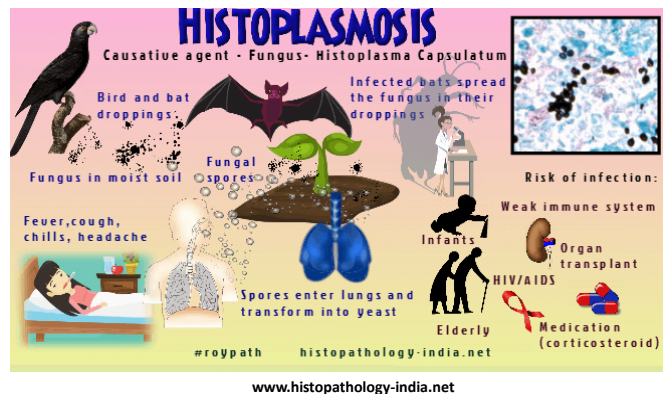


*Map has been adapted from CDC. Borders are approximate.

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Histoplasmosis

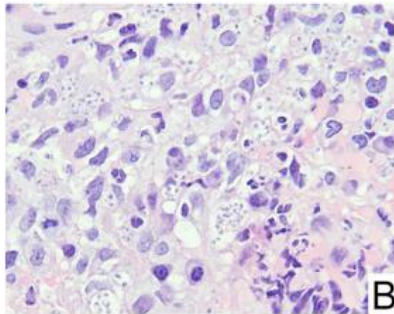
- *Histoplasma capsulatum*
- Common systemic fungal infection in the United States
- Common in the Ohio-Mississippi River Valley
- Spores are inhaled from bird or bat droppings
 - Mold → yeast in body
- Majority of the cases produce little if any symptoms
- Radiographs reveal calcification of the hilar lymph nodes
- Oral lesions present as painful solitary ulcerations



Histoplasmosis – Oral Lesion with Histology

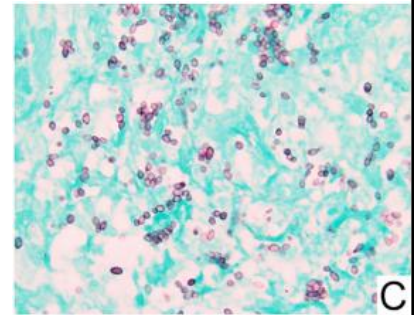


Deep chronic ulcer with raised borders on the dorsal surface of the tongue



Small circular fungal organisms surrounded by chronic granulomatous inflammation.

GMS stain used to highlight fungal organ



International Journal of Surgery Case Reports 93 (2022) California Northstate University College of Dental Medicine Head and Neck Pathology (2021) 15:682–690

Histoplasmosis

Calcification of lymph nodes in chest cavity



www.medicalnewstoday.com/articles/pictures-of-histoplasmosis-in-the-lungs#pictures
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Blastomycosis

- *Blastomyces dermatitidis*
- Lesions of oral cavity, skin, lungs, CNS, bone, and GU tract
- Male predilection
 - 9:1 ratio
 - Occurs in men involved in outdoor activities
 - Common to the states surrounding the Great Lakes and Canada
 - Overlaps with histoplasmosis in certain locations in the U.S.
- Spores are inhaled especially after rainfall

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Blastomycosis- Oral Lesions with Histology

Non-healing extraction site showing a defect. The affected tissue has an ulcerated, granular, and slightly erythematous appearance.

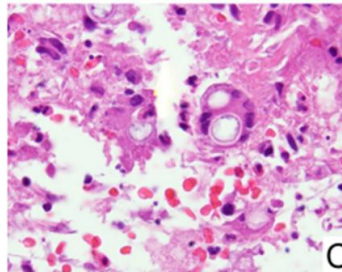


A



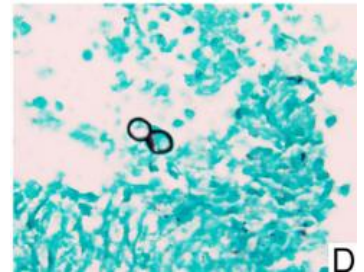
B

Radiographs show a portion of the non-healing extraction site. Note the widening of the PDL surrounding the root of the canine next to the extraction site.



C

Large fungal organism with a double-refractile capsule surrounded being engulf by a phagocytic cell within fibrous connective tissue containing necrosis and other inflammatory cells.



D

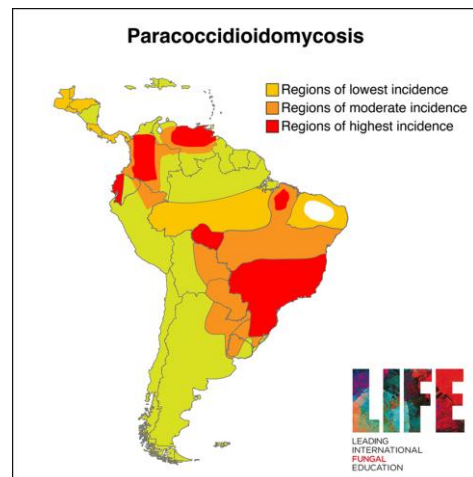
GMS stain highlight the fungal organism.

Head and Neck Pathology (2021) 15:682–690

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Paracoccidiomycosis

- *Paracoccidioides brasiliensis*
- Deep fungal infection seen in South and Central America
- “South American Blastomycosis”
- Male predilection
- 15:1
- Middle-aged agricultural workers
- Beta-estradiol prevents hyphae from transforming to the yeast form in women



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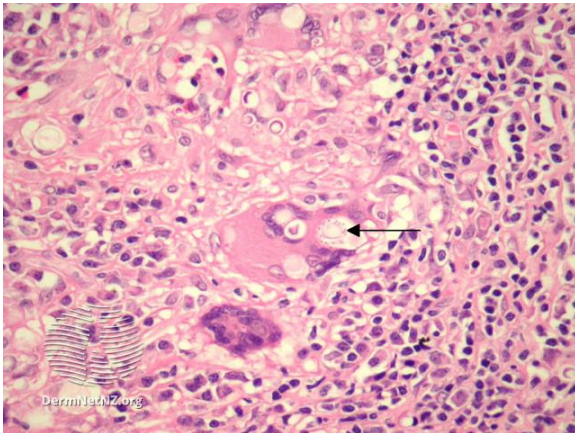
Paracoccidioidomycosis - Oral Lesion



Ulcerated and granular appearance to the tissue involving the anterior mandibular alveolar ridge and floor of mouth.

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Journal of Surgical and Clinical Dentistry Vol. 5, n. 1, pp. 12-15 (Apr - Jun 2015)

Paracoccidioidomycosis - Histology



Chronic granulomatous inflammation surrounding the large fungal organisms



A parent cell with multiple budding daughter cells. The budding organism resembles "Mickey Mouse" ears or the spokes of a ships steering wheel. These histology descriptions a characteristic microscopic features associated with paracoccidioidomycosis infections.

California Northstate University College of Dental Medicine <https://phil.cdc.gov/Details.aspx?pid=527>

Coccidioidomycosis

- *Coccidioides immitis*
- Common in the Southwestern U.S., Mexico, and Central America
- Grows in the desert soil
- Most infections are asymptomatic
- 40% experience flu-like symptoms
- Chronic variant mimics tuberculosis
- Disseminated variant is seen in the immunocompromised patients

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Coccidioidomycosis

- San Joaquin Valley Fever
 - Coccidioidomycosis
 - Erythema Multiforme or erythema nodosum
 - Erythema multiforme: Ulcerative oral lesions and red lesions of the skin
 - Erythema nodosum: Painful red nodules involving the subcutaneous connective tissue

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Coccidioidomycosis



Mendez, Luis A et al. "Ulcerated Lesion of the Tongue as Manifestation of Systemic Coccidioidomycosis." Case reports in medicine vol. 2017 (2017)

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Skin Lesions Associated With San Joaquin Valley Fever



Erythema Nodosum



Erythema Multiforme

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ESTOP Conferences

San Joaquin Valley Fever

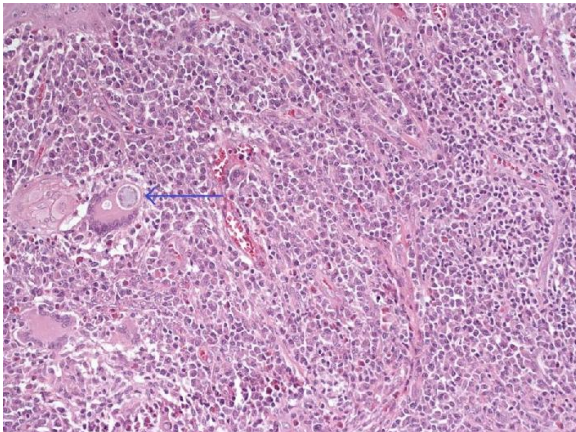


youtu.be/tyHqQxltEoM?si=rGqn-_zc8sBkL-GT&t=5

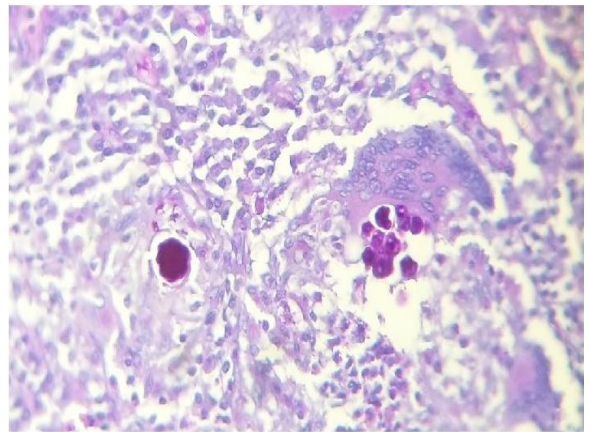
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Coccidioidomycosis



Chronic granulomatous inflammation surrounding a spherule filled with endospores (identified by blue arrow).



PAS stain used to highlight the fungal organisms

Mendez, Luis A et al. "Ulcerated Lesion of the Tongue as Manifestation of Systemic Coccidioidomycosis." Case reports in medicine vol. 2017 (2017)

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Zygomycosis (Mucormycosis)

- Opportunistic infection
- Organisms are found in bread molds, decaying fruits or vegetables
- Found in uncontrolled insulin-dependent diabetics and other immunocompromised patients
- Symptoms: nasal obstruction, facial swelling, and visual disturbances

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Mucormycosis – Palatal Perforation

The fungal organism can invade blood vessels leading to thrombus formation causing necrosis of the bone and soft tissue.

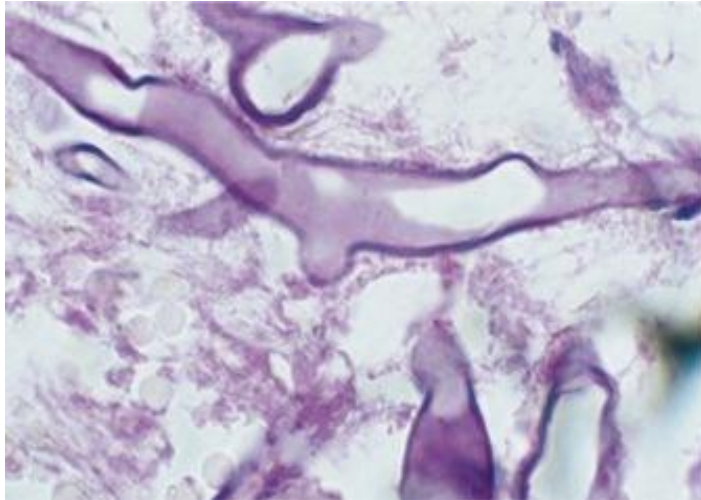


Courtesy of Oral Pathology Lab Inc.

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Mucormycosis - Histology

Microscopic examination will show, large, non-septate fungal hyphae which branch at 90-degree angles.



Contemporary Oral and Maxillofacial Pathology 2nd Edition
California Northstate University College of Dental Medicine

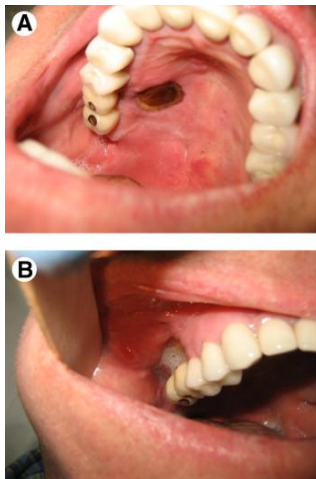
Aspergillosis

- Transmitted by spores released in the air and inhaled
- Nosocomial infection
- Fungal disease that can be non-invasive or invasive
- **Non-invasive**
 - Allergic fungal sinusitis
- **Invasive**
 - Uncontrolled diabetes mellitus, HIV+ patients, patient with leukemia, and those on high daily doses of corticosteroids
 - Can cause disseminated disease

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Invasive Aspergillosis

Necrotic area affecting both the hard palate and buccal alveolar bone.

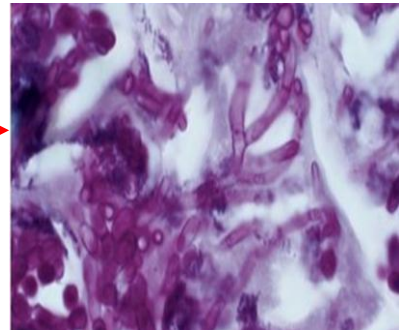
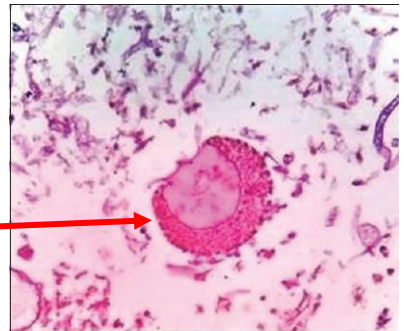


Fuqua, Thomas H Jr et al. "Primary invasive oral aspergillosis: an updated literature review." *Journal of oral and maxillofacial surgery : official journal of the American Association of Oral and Maxillofacial Surgeons* vol. 68,10 (2010): 2557-63

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Aspergillosis

- Histology
 - Fungus ball/fruiting body
 - Non-invasive
 - Invades small blood vessels
 - Invasive
 - Septate hyphae
 - Branching at acute angles
- Treatment depends on clinical presentation



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Cryptococcosis

- Most life-threatening to immunocompromised patients
 - *Cryptococcus neoformans*
- Associated with pigeon droppings
- Acquired by inhalation
- Patients present with flu-like symptoms
- Disseminated disease
 - Meninges, skin, bone, and prostate gland
 - Meningitis: headache, vomiting, neck stiffness, and fever

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Cryptococcosis – Skin and Oral Manifestations



Disseminated disease showing erythematous plaques located on the skin.

Dinato, Sandra Lopes Mattos et al. "Disseminated cutaneous cryptococcosis in a patient with AIDS." *Revista do Instituto de Medicina Tropical de Sao Paulo* 48 6 (2006): 353-8 .

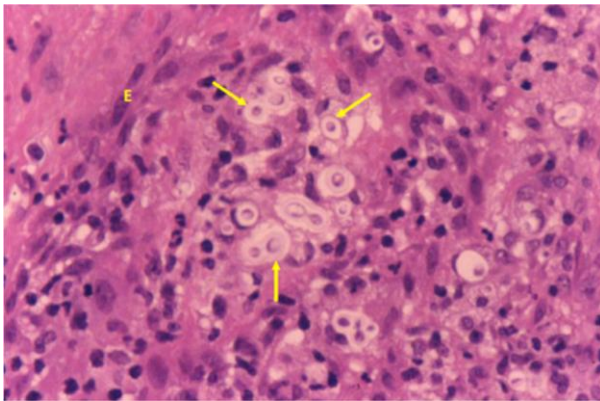


Granular, erythematous growths affecting the oral mucosa.

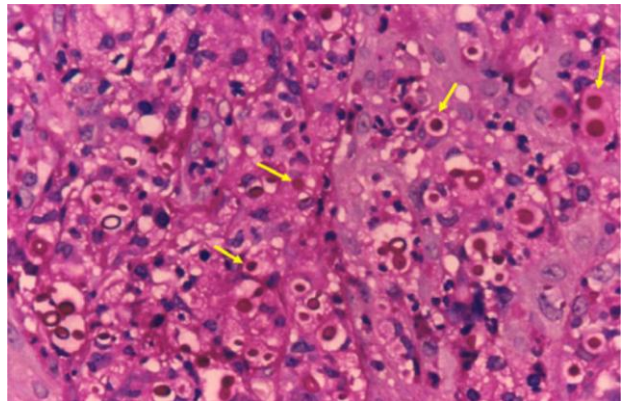
Delgado, Wilson A.. "Oral Cryptococcosis." (2017).

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Cryptococcosis - Histology



Fungal organisms surrounded by chronic granulomatous inflammation



PAS stain highlighting fungal organism

Delgado, Wilson A.. "Oral Cryptococcosis." (2017).
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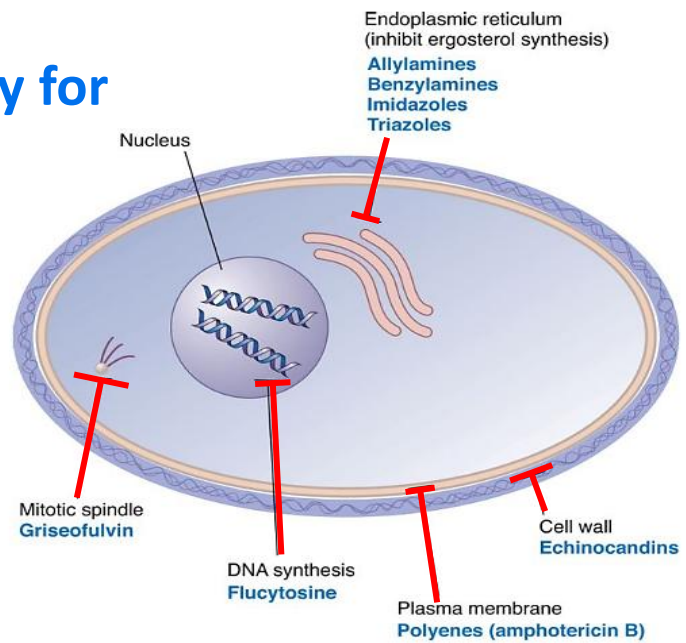
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Pharmacological Management

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Therapeutic strategy for Anti-fungal drugs



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Treatment of oral *Candidiasis*

- Clotrimazole 10 mg oral troches
- Nystatin oral suspension (100, 000 U/ml).
- Miconazole mucoadhesive buccal tablet 50 mg.

HIV positive or in case of severe infections: oral Fluconazole. IV amphotericin B with or without flucytosine may be considered.

Pregnant patients: Avoid oral azoles. Use topical therapies like clotrimazole troches or miconazole buccal tablets (50 mg).

TABLE 34-2 Treatment Strategies for Oral Candidiasis (Oral Thrush)

Agent	Formulation	Dosage
Clotrimazole	Oral troches [‡]	10 mg orally, 5x/day, for 14 d
Nystatin	Oral suspension	4-6 mL (400,000-600,000 U), 4x/day until resolution
Miconazole	Buccal tablets [‡]	One 50-mg tab per day, placed on the gingiva of the canine fossa above tooth #8 and #9
Itraconazole [†]	Oral	200 mg daily, for 7-14 d
Fluconazole [†]	Oral	200 mg on day1, then 100 mg qd, for 14 d
Posaconazole [†]	Oral suspension	100 mg (2.5 mL) bid on day1, then 100 mg (2.5 mL) qd, for 13 d
Caspofungin [†]	IV solution	50 mg once/day for 7-14 d, or longer if more severe

*Doses are for adult patients.

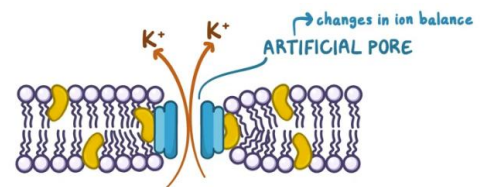
[†]For more serious or extensive infections.

[‡]Remain in place until dissolved.

Polyenes

Example: Nystatin oral suspension and Amphotericin B

MOA: Polyenes bind to ergosterol in fungal cell membranes, creating ion channels (pores). Pores disrupt of membrane function leading to leakage of small molecules and cell death.



Pharmacokinetics:

1- Nystatin: Not absorbed from the GI tract. Used as topical, intravaginal or oral to treat infections of cutaneous or mucosal surfaces.

2- Amphotericin: poor GI absorption. Administered via IV for severe fungal infections. **Adverse effect:** nephrotoxicity

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Azoles

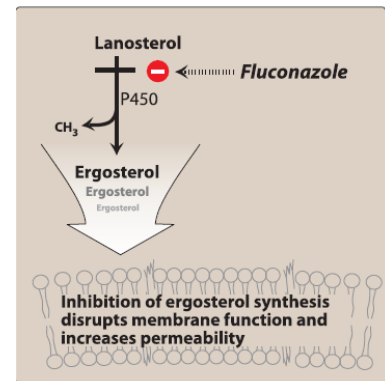
Examples include

- Imidazole (Ketoconazole and miconazole).
- Triazoles (fluconazole and Itraconazole).

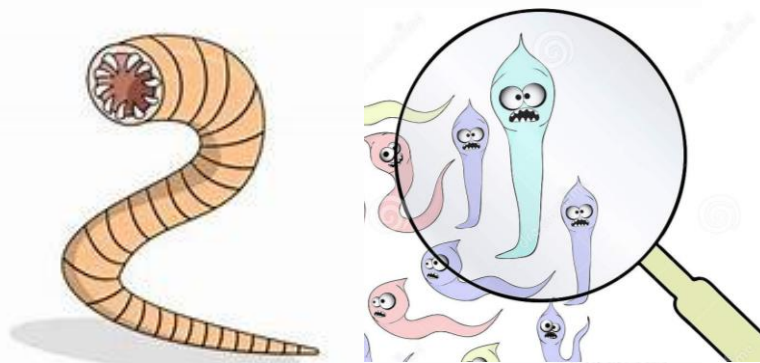
MOA: Inhibit 14- α demethylase which blocks the demethylation of lanosterol to ergosterol.
Inhibition of ergosterol biosynthesis disrupts fungal membrane structure and function.

Azoles are teratogenic.

Azoles are CYP 450 inhibitors



PARASITE



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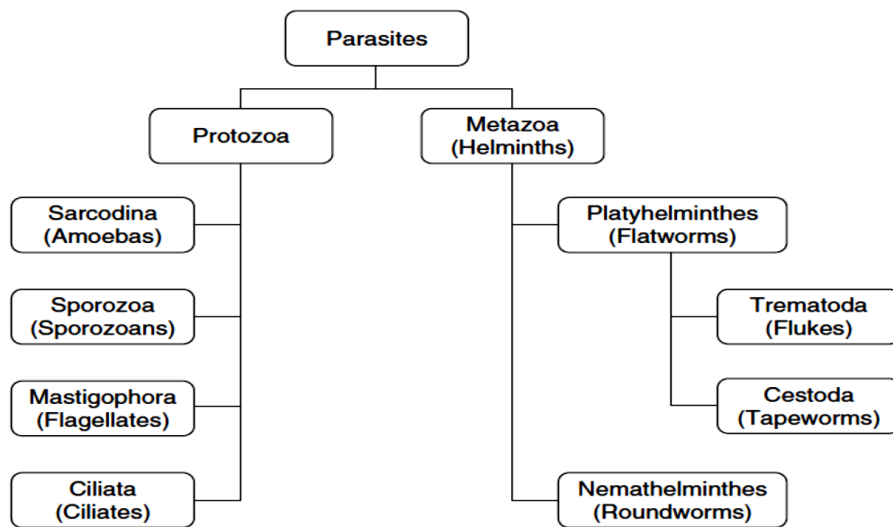


FIGURE 18-15. Phylogeny of parasites.

Reproduced, with permission, from Levinson W. *Medical Microbiology and Immunology*, 8th ed. New York: McGraw-Hill, 2004.

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Entamoeba gingivalis* and *Trichomonas tenax

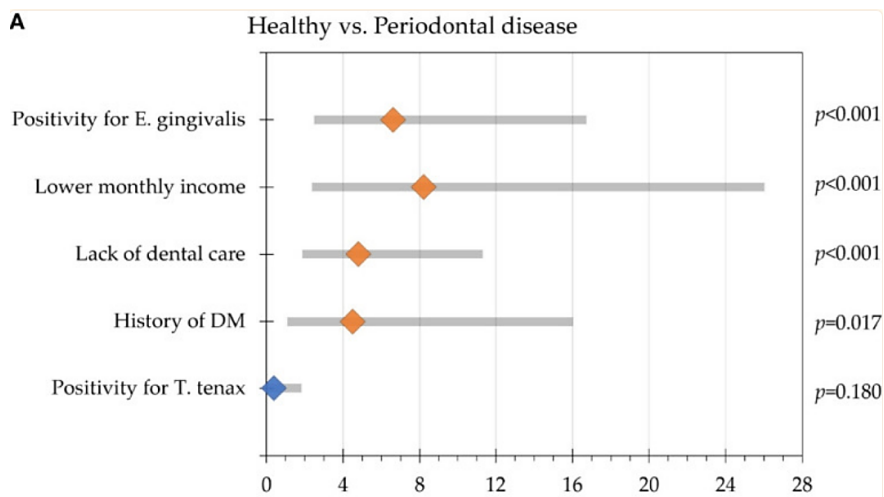
Host: Humans and animals

Transmission: Oral-oral route with human or animals and shared utensils

***E. gingivalis*:** can be associated with periodontitis. Prevalence: 88.9% among periodontitis patients, 84.9% among gingivitis patients.

***T. tenax*:** Prevalence 25.6% among periodontitis patients, 5.7% among gingivitis patients. Can cause trichomoniasis (a common STI).

T. gingivalis is correlated with periodontal disease



Oral Colonization by *Entamoeba gingivalis* and *Trichomonas tenax*: A PCR-Based Study in Health, Gingivitis, and Periodontitis - PMC (nih.gov) California Northstate University College of Dental Medicine

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Plasmodium sp.

PROTOZOAN	DISEASE	TRANSMISSION	MORPHOLOGY	TREATMENT
<i>Plasmodium vivax</i>	Malaria	Female	Sporozoite.	Chloroquine
<i>Plasmodium ovale</i>	(influenza-like	mosquitoes	Merozoite (liver).	Mefloquine
<i>Plasmodium</i>	onset):	(<i>Anopheles</i>)	Hypnozoite (latent).	Primaquine
<i>malariae</i>	Fever	Infected blood	Causes lysis of erythrocytes.	
<i>Plasmodium</i>	Headache	products		
<i>falciparum</i>	Anemia			
(most fatal)	Splenomegaly			

[Plasmodium species \(Malaria\)](#)
[| Osmosis](#)

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Treatment strategies

Chloroquine: Anti-malarial, for intestinal amebiasis & anti-inflammatory.

- Can increase risk for acute hemolysis in patients with G6PD.
- Can cause drug-induced pigmentation of the gingiva, palate; may occur due to stimulation of melanin synthesis, drug accumulation.



[\(PDF\) Hyperpigmentation of hard palate induced by chloroquine therapy \(researchgate.net\)](#)

Metronidazole is a broad spectrum antiprotozoal and antibacterial.

- Patients should not mix with alcohol due to disulfiram-like reaction.

Big Picture

- Learn the distinctive features for fungi and difference between yeast vs mold
- Study the different categories of fungal diseases and treatment strategies.
- Focus on dentally relevant protozoa and their treatment.
- Know the deep fungal infection based on where they are found, susceptible populations, and clinical symptoms.
 - Palatal perforations and DDX of oral cancer!!!!

Resources

Rajendra Santosh, Arvind Babu et al. "Fungal Infections of Oral Cavity: Diagnosis, Management, and Association with COVID-19." SN comprehensive clinical medicine vol. 3,6 (2021): 1373-1384.

